

NEXUS

AI-Powered Supply Chain Intelligence, Risk Prediction & Optimization Platform

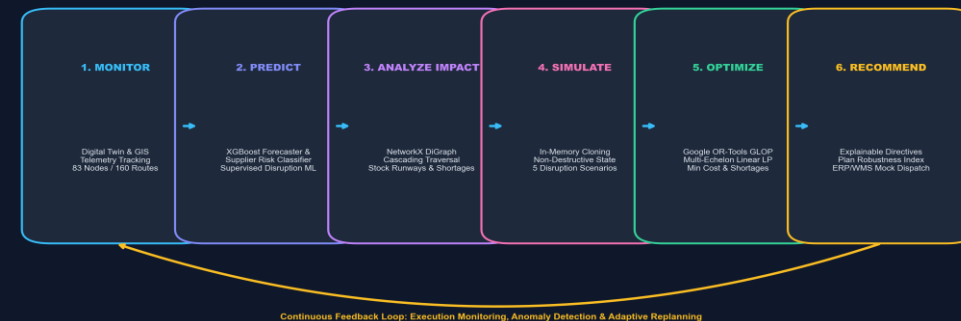
MONITOR → PREDICT → ANALYZE IMPACT → SIMULATE → OPTIMIZE → RECOMMEND

PROJECT PRESENTATION & DEFENSE

- **Candidate:** Shreyansh Uttam (22BCE10738)
- **Program:** B.Tech Computer Science & Engineering (AI & ML)
- **School:** School of Computing Science and Engineering (SCSE)
- **Institution:** VIT Bhopal University, Madhya Pradesh
- **Focus Area:** Operations Research, Machine Learning & Digital Twins

The NEXUS Closed-Loop Decision Intelligence Cycle

Automated Transition from Disruption Ingestion to Quantified Mitigation Directives



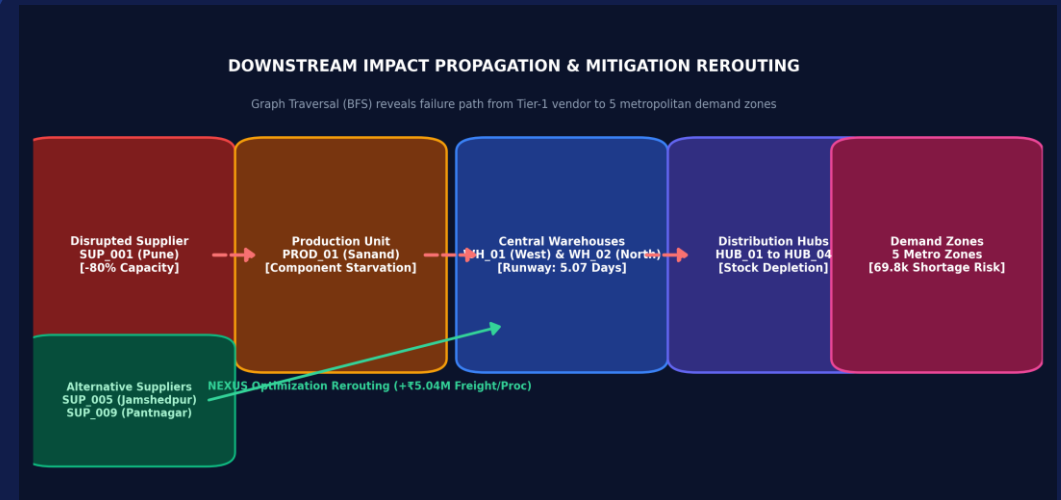
The Supply Chain Fragility Problem

Modern multi-echelon supply networks suffer from fragmented data, delayed analysis, and rigid static contracts.

TRADITIONAL DECISION BOTTLENECKS

- **The Traditional Workflow:** Raw Data → Spreadsheets / Static ERP Reports → Manual Human Analysis → Delayed Decisions.
- **Demand Uncertainty & Bullwhip Effect:** Small shifts in consumer velocity amplify upstream into massive order swings, causing stockouts or inventory glut.
- **Multi-Tier Blindspots:** Enterprises lack visibility past Tier-1 suppliers; Tier-2 subassembly failures trigger sudden production halts.
- **Rigid Single-Sourcing:** Fixed contracts prevent dynamic rerouting; when a primary vendor collapses, downstream facilities face immediate stockouts.
- **High Contractual Penalties:** Unmitigated stockouts accumulate steep contractual SLA breach fees (e.g. INR 350 per unmet unit).
- **Disconnected Systems:** Forecasting, inventory, and logistics exist in separate silos without an automated closed-loop feedback engine.

DISRUPTION CASCADE & THE CORE NEED



THE ENGINEERING NEED:

A unified decision intelligence platform that transitions supply chain management from reactive firefighting to automated, mathematically proven mitigation before stockouts materialize.

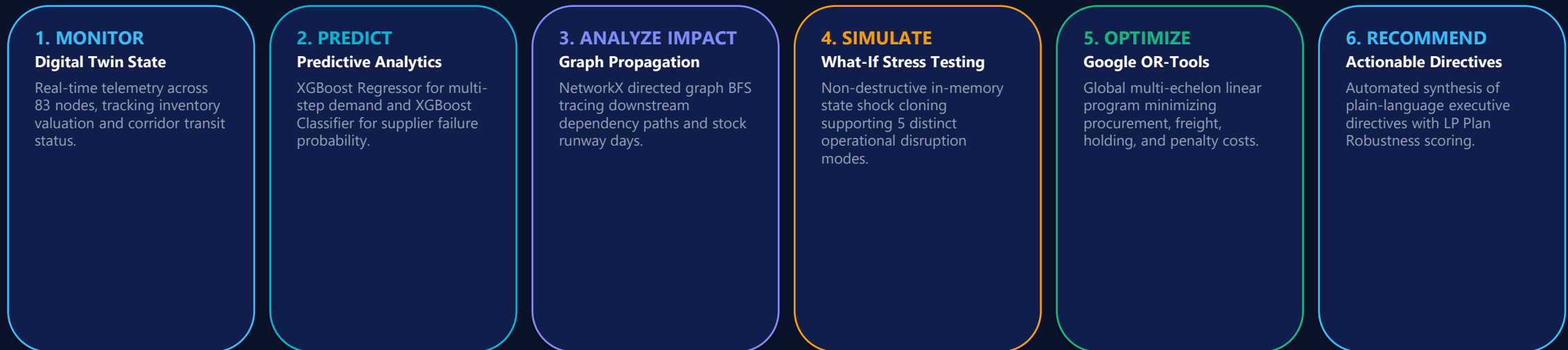
From Monitoring to Decision — The Closed Loop

NEXUS unifies monitoring, machine learning, graph analysis, simulation, and optimization into a single automated pipeline.

NEXUS CLOSED-LOOP DECISION INTELLIGENCE ARCHITECTURE

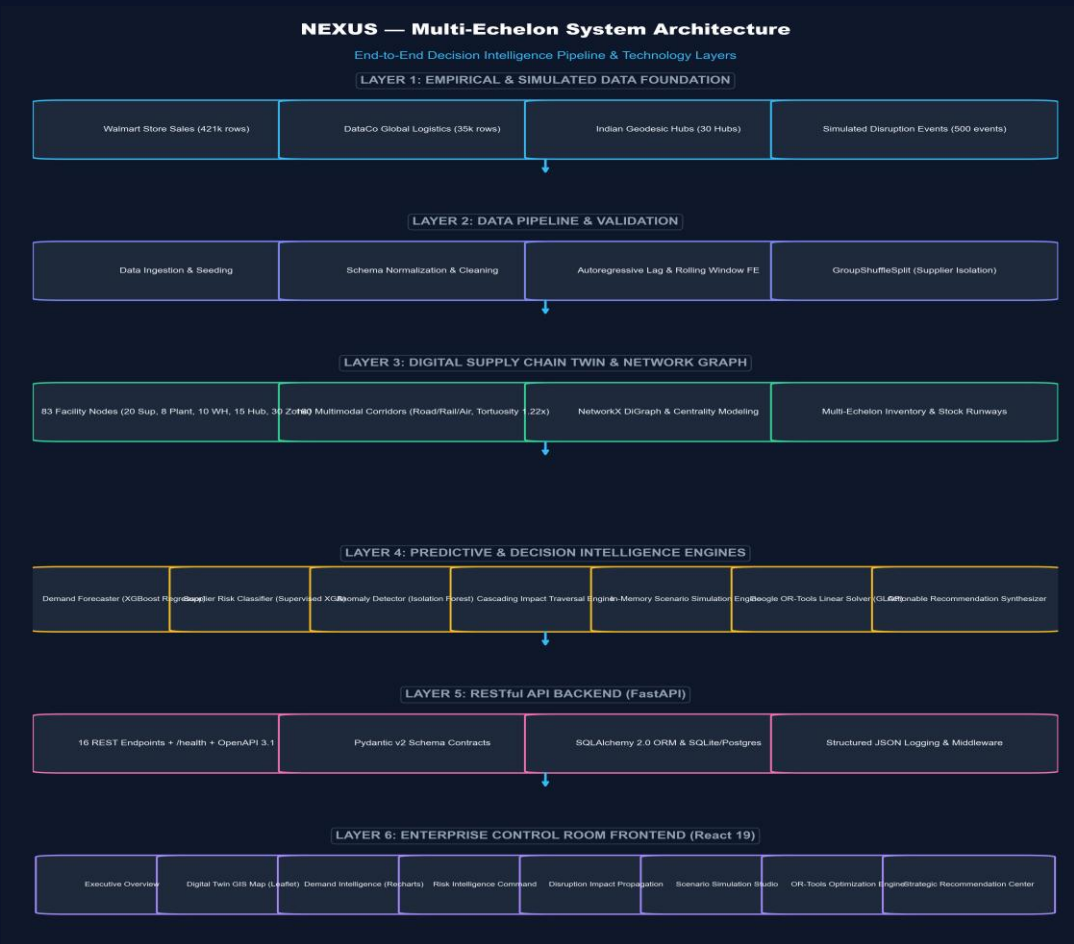


Closed feedback loop converting empirical sensing into mathematically verified operational allocations



How NEXUS Works — End-to-End Architecture

A decoupled, seven-layer engineering architecture connecting empirical datasets, ML pipelines, optimization, and React UI.



SEVEN-LAYER DECOUPLED STACK

- **Layer 1: Ingestion & Provenance:** Walmart sales (421k rows), DataCo shipments (35k rows), and Indian hub geodesy.
- **Layer 2: Preprocessing & Validation:** Strict Pydantic v2 schemas, Parquet serialization, and 14-lag feature engineering.
- **Layer 3: Relational Persistence:** SQLite 3 database (14 tables, 50k orders, 500 inventory items, PostgreSQL DDL ready).
- **Layer 4: Machine Learning & Analytics:** XGBoost Regressor, XGBoost Risk Classifier, and Isolation Forest anomaly detector.
- **Layer 5: Graph & Simulation Engine:** NetworkX multi-echelon directed graph (BFS cascade) and in-memory copy-on-write ScenarioEngine.
- **Layer 6: Operations Research:** Google OR-Tools GLOP linear solver enforcing multi-echelon flow, capacity, and demand satisfaction.
- **Layer 7: Presentation & Control Room:** FastAPI (20 REST endpoints) driving a modern React 19 TypeScript control room.

Data Foundation — Empirical vs Simulated Data

NEXUS combines real-world empirical time-series and shipment records with a calibrated synthetic digital twin.

Dataset Source	Nature of Data	Verified Volume / Scope	Target Analytical Purpose in NEXUS
Walmart Store Sales	Public Empirical (Kaggle Research)	421,570 weekly historical records 45 stores, 81 depts, 2.5+ years horizon	Multi-step weekly demand forecasting, autoregressive lags, calendar seasonality
DataCo Global Supply Chain	Public Empirical (Mendeley Data CC BY 4.0)	35,000+ representative shipments 180,519 order corpus, real transit delays	Supervised supplier disruption risk, lead-time variability, delivery failure modeling
Indian Logistics Digital Twin	Calibrated Synthetic (Reproducible Topology)	83 facility nodes, 160 corridors, 50,000 orders, 500 inventory items	Multi-echelon network topology, geodesy, impact propagation, OR-Tools optimization

EMPIRICAL GROUNDING (REAL DATA)

- **Real Sales Volatility:** Captures genuine retail demand spikes, holiday surges, and trend seasonality that synthetic sine-waves cannot reproduce.
- **Real Delivery Delays:** Trained on authentic international freight delay distributions where actual transit days exceeded scheduled days.

ACADEMIC HONESTY (SIMULATED TWIN)

- **Simulated Indian Topology:** Real corporate facility BOMs and internal supplier networks are strictly confidential.
- **Calibrated Distributions:** NEXUS maps empirical Walmart and DataCo distributions onto realistic Indian hubs (Delhi, Mumbai, Pune, Sanand, Bengaluru).

From Raw Data to Decision-Ready Intelligence

Systematic feature engineering, temporal shift guards, and in-memory state representations.

1. DEMAND TIME-SERIES ENGINEERING

- **Autoregressive Lags:** Lags $t-1$, $t-2$, $t-4$, $t-8$ capturing cyclical ordering patterns.
- **Rolling Window Stats:** 4-week and 8-week moving averages (μ) and 4-week rolling volatility (σ).
- **Temporal Guard:** All rolling metrics strictly shifted by 1 period ($\text{shift}(1)$) to prevent lookahead data leakage.
- **Calendar Seasonality:** Deterministic trend index, ISO week of year, month, and holiday flags.

2. SUPPLIER RISK FEATURE EXTRACTION

- **Lead-Time Volatility:** Standard deviation of delivery lead-time (primary collapse predictor).
- **Operational Metrics:** Historical delay frequency, average late days, and on-time fulfillment rate.
- **Capacity Strain:** Plant capacity utilization ratio and quality assurance defect rate.
- **Stochastic Shock:** Latent operational stress modeling unannounced industrial failures.

3. SPATIAL GEODESY & CORRIDORS

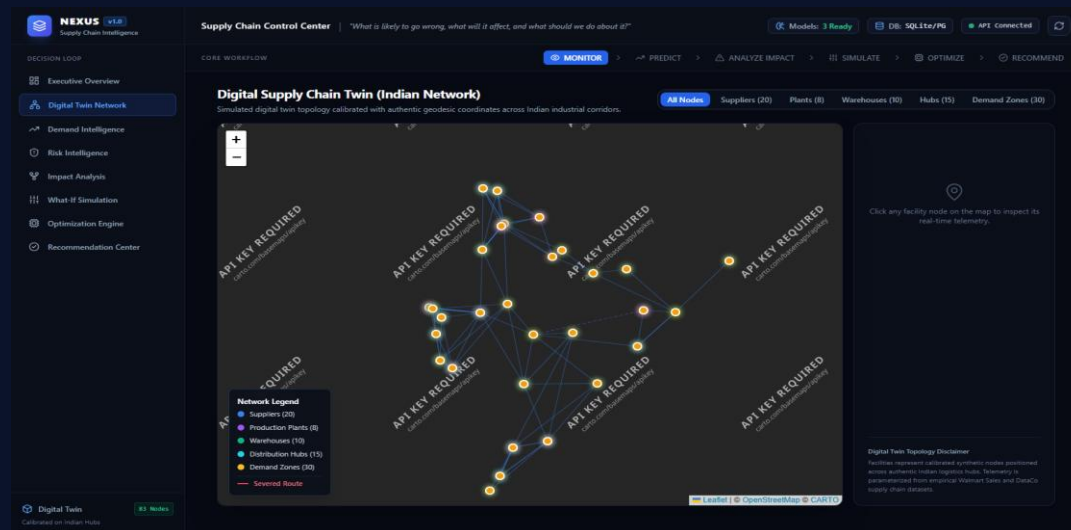
- **Geocoded Facilities:** 83 exact geographic coordinates across canonical Indian logistics corridors.
- **Highway Tortuosity:** Haversine distance multiplied by 1.22x tortuosity factor for authentic road transit km.
- **Multimodal Modes:** Differentiated transit speeds and costs for National Expressways vs Dedicated Rail Freight.
- **Corridor Capacity:** Throughput bottlenecks and maximum lane vehicle volumes.

4. OPERATIONAL RUNWAY & SHOCK STATE

- **Stock Runway Metric:** $\text{Runway (Days)} = \text{Current Inventory} / \text{Daily Consumption Burn Rate}$.
- **Buffer Depletion:** Pinpoints exact calendar day of catastrophic warehouse stockout.
- **In-Memory State Cloning:** Deep copy of `SimulationState` instantiates non-destructive shock test.
- **Allocation Feasibility:** Calculates net gross capacity surplus/deficit across all network echelons.

The NEXUS Digital Twin — Calibrated Multi-Echelon Network

An 83-node, 160-corridor digital representation spanning five operational supply chain tiers.



VERIFIED DIGITAL TWIN METRICS

- **Total Facilities:** 83 geocoded nodes across canonical Indian logistics corridors.
- **20 Suppliers:** Tier-1 & Tier-2 automotive, metallurgy, polymer, and precision parts.
- **8 Production Units:** OEM manufacturing & assembly plants (Sanand, Pune, Gurgaon, Sriperumbudur).
- **10 Central Warehouses:** Regional distribution centers tracking 500 SKUs with INR 45.92M valuation.
- **15 Distribution Hubs:** Intermediate multimodal transshipment and sorting facilities.
- **30 Demand Zones:** Tier-1/Tier-2 consumption centers representing 153.6k units of demand.
- **160 Corridors:** Active freight transit lanes calibrated with authentic transit hours.

Intelligence Layer — Forecasting, Risk & Anomaly Detection

Three specialized analytical models addressing distinct operational supply chain failure modes.

1. DEMAND FORECASTING

Architecture: XGBoost Regressor

- **Objective:** Predict multi-step demand velocity over a 4-week horizon to prevent bullwhip distortion.
- **Input Vector:** 14 engineered features: autoregressive lags (t-1 to t-8), rolling mean/std (4W/8W), calendar seasonality.
- **Algorithm:** Gradient Boosted Trees (150 estimators, max depth 4, learning rate 0.05, subsample 0.85).
- **Output:** Weekly expected demand quantity with multi-horizon confidence intervals.

2. SUPPLIER DISRUPTION RISK

Architecture: XGBoost Classifier

- **Objective:** Predict the probability of vendor SLA breach or operational failure before stockouts occur.
- **Input Vector:** 8 operational variance metrics: lead-time std dev, nominal lead-time, delay frequency, OTR, quality score.
- **Algorithm:** Supervised classification with `scale_pos_weight=1.5` and tree depth 3.
- **Output:** Disruption probability score [0.0, 1.0], risk tier (Low/Med/High/Critical), and explainable feature weights.

3. OPERATIONAL ANOMALY DETECTION

Architecture: Isolation Forest

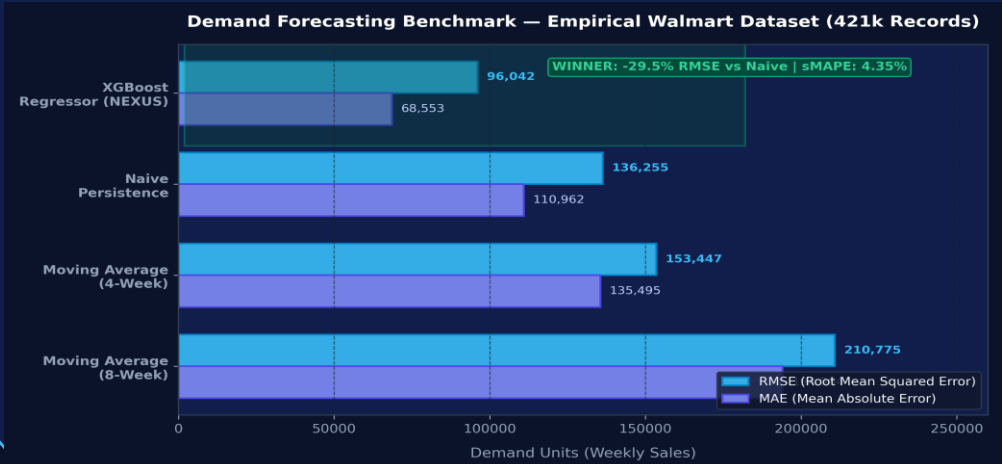
- **Objective:** Unsupervised real-time detection of sudden demand surges and unprecedented transit delays.
- **Input Vector:** Continuous transaction streams: order quantities, delivery deviations, and velocity spikes.
- **Algorithm:** Isolation Forest (100 estimators, contamination rate $\alpha = 0.03$).
- **Output:** Normalized anomaly score and classification taxonomy (UNUSUAL_DEMAND_SPIKE, TRANSIT_DELAY_OUTLIER).

Empirical Model Evaluation & Benchmark Comparison

Rigorous out-of-sample benchmarking against industry baseline algorithms on strictly held-out test data.

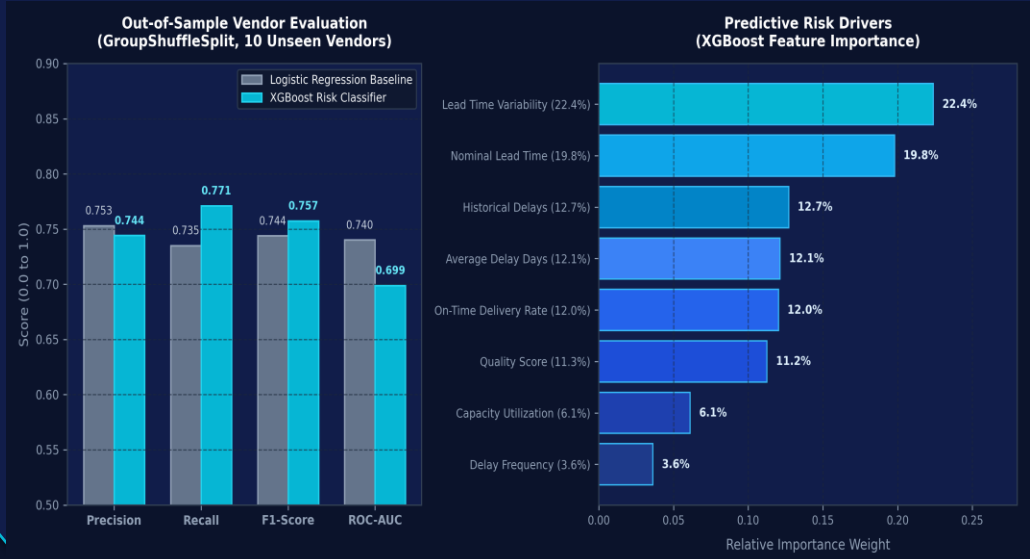
DEMAND FORECASTING BENCHMARK

Model Architecture	MAE	RMSE	sMAPE	Rank
Naive Persistence	110,961.63	136,255.10	7.20%	2
Moving Avg (4-Week)	135,495.41	153,446.84	8.42%	3
Moving Avg (8-Week)	194,063.88	210,775.27	11.80%	4
XGBoost Regressor (NEXUS)	68,553.14	96,042.50	4.35%	1 (Winner)



SUPPLIER RISK OUT-OF-SAMPLE BENCHMARK

Model Architecture	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression Baseline	0.7531	0.7349	0.7439	0.7401
XGBoost Classifier (NEXUS)	0.7442	0.7711	0.7574	0.6988



The NEXUS Decision Matrix — From Signal to Action

Prediction alone does not make decisions; NEXUS pairs signals with downstream impact and mathematical optimization.

Operational Signal	Detection Mechanism	Downstream Network Impact	NEXUS Prescriptive Decision
Demand Spike / Surge	XGBoost Regressor (multi-step forecast)	Depletes warehouse safety buffer; stockout risk within 14 days	Dynamically scale production quotas & increase supplier procurement allocations
Supplier Capacity Loss	Supervised Risk Classifier + ScenarioEngine	Upstream component starvation; Tier-1 assembly line halt	Shift sourcing allocation to qualified secondary vendors (e.g. SUP_005, SUP_009)
Route Delay / Blockage	NetworkX Graph Centrality & Geodesy Engine	Transit time extended by 2-5 days; late delivery SLA penalties	Reroute freight via alternative multimodal rail/expressway corridors
Inventory Runway Depletion	Runway Calculus (Stock / Daily Burn Rate)	Imminent stockout (< 7 days runway) at regional central warehouse	Trigger rapid inter-warehouse safety stock transfers via rail corridors
Multi-Node System Shock	Scenario Simulation Engine (copy-on-write state)	Cascading multi-tier deficit across multiple metropolitan demand zones	Execute Google OR-Tools multi-echelon linear program to find global cost-optimal allocation

THE NEXUS DECISION PRINCIPLE

Prediction is only an input, not a solution.

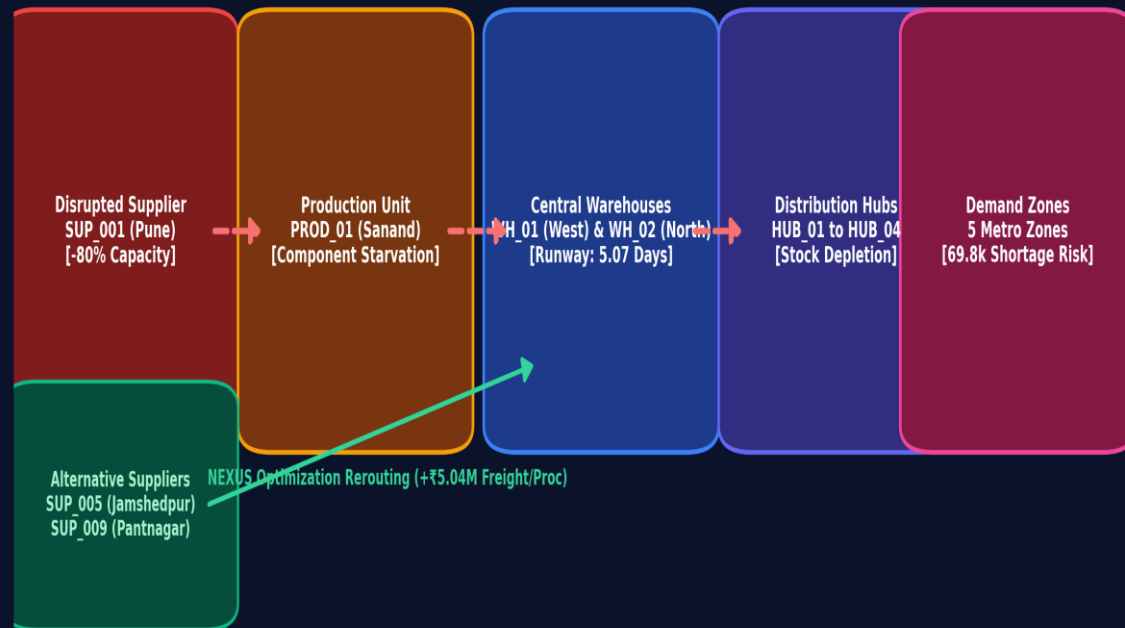
A prediction of 80% supplier risk does not tell a manager how many units to reroute, which alternative vendor has capacity, or what the freight cost tradeoff will be. NEXUS bridges this gap by combining Prediction + Network Impact + Scenario Simulation + Mathematical Optimization into concrete, actionable decisions.

Impact Propagation — What Happens When a Node Fails?

NetworkX directed graph traversal (BFS) traces the ripple effect of upstream shocks across all downstream tiers.

DOWNSTREAM IMPACT PROPAGATION & MITIGATION REROUTING

Graph Traversal (BFS) reveals failure path from Tier-1 vendor to 5 metropolitan demand zones



GRAPH TRAVERSAL & RUNWAY CALCULUS

- **Multi-Tier Graph Traversal:** NetworkX directed graph executes Breadth-First Search (BFS) starting from the disrupted supplier node down to final demand zones.
- **Component Dependency Mapping:** Directly maps disrupted supplier parts to downstream Bill-of-Materials (PROD_ITEM_003 automotive subassembly).
- **Warehouse Runway Calculus:** Runway Days = Current Stock / Daily Burn Rate. For Western Warehouse (WH_01): 1,420 units / 280 units/day = 5.07 Days Runway.
- **Pinpointing Stockout Dates:** Pinpoints exact day of stockout: WH_01 depletes completely on Day 6 without intervention.
- **Service Level Degradation:** Network-wide on-time fulfillment projected to plummet from 98.0% baseline down to 54.57%.
- **Exposed Demand Volume:** 69,843.2 units of unmet demand exposed across 5 metropolitan consumption zones (Mumbai, Pune, Delhi NCR, Jaipur, Ahmedabad).
- **Automated Mitigation Search:** Graph engine immediately queries active secondary suppliers with matching product categories and surplus capacity.

Scenario Simulation — Non-Destructive In-Memory Testing

NEXUS

v1.0

Supply Chain Intelligence

DECISION LOOP

Executive Overview

Digital Twin Network

Demand Intelligence

Risk Intelligence

Impact Analysis

What-If Simulation

Optimization Engine

Recommendation Center

Supply Chain Control Center

"What is likely to go wrong, what will it affect, and what should we do about it?"

Models: 3 Ready

DB: SQLite/PG

API: Connected

CORE WORKFLOW

MONITOR

PREDICT

ANALYZE IMPACT

SIMULATE

OPTIMIZE

RECOMMEND

What-If Scenario Simulation Builder

State Isolation Active (Non-destructive)

Scenario Shock Builder

Supplier Cut

Route Severance

Demand Surge

WH Shutdown

Combined Compound

Supplier Node:

Capacity Cut:

80%

Disruption Duration:

10 Days

SUP_001 - Tata AutoComp Components

Execute Digital Twin Simulation

Before vs After Shock State Comparison

Scenario: SUPPLIER_FAILURE

NOMINAL BASE STATE (BEFORE)

Healthy Network

Total Consumer Demand:

153,750 units

Active Supplier Capacity:

314,500 units

Net Capacity Reserve:

+160,750 units

Severed Routes:

0 corridors

Nominal Service Level:

98.5%

SIMULATED SHOCK STATE (AFTER)

Shock Applied

Total Consumer Demand:

1,53,750 units

Available Supplier Capacity:

3,14,500 units

Compromised Suppliers:

1 vendor(s)

Severed Routes:

0 corridor(s)

Net Capacity Balance:

1,60,750 units

Shock state successfully captured in digital twin memory.

Proceed to Optimization to compute the mathematically optimal multi-echelon response using Google OR-Tools.

Solve with OR-Tools

Digital Twin

83 Nodes

Calibrated on Indian Hubs

FIVE VERIFIED DISRUPTION SCENARIOS

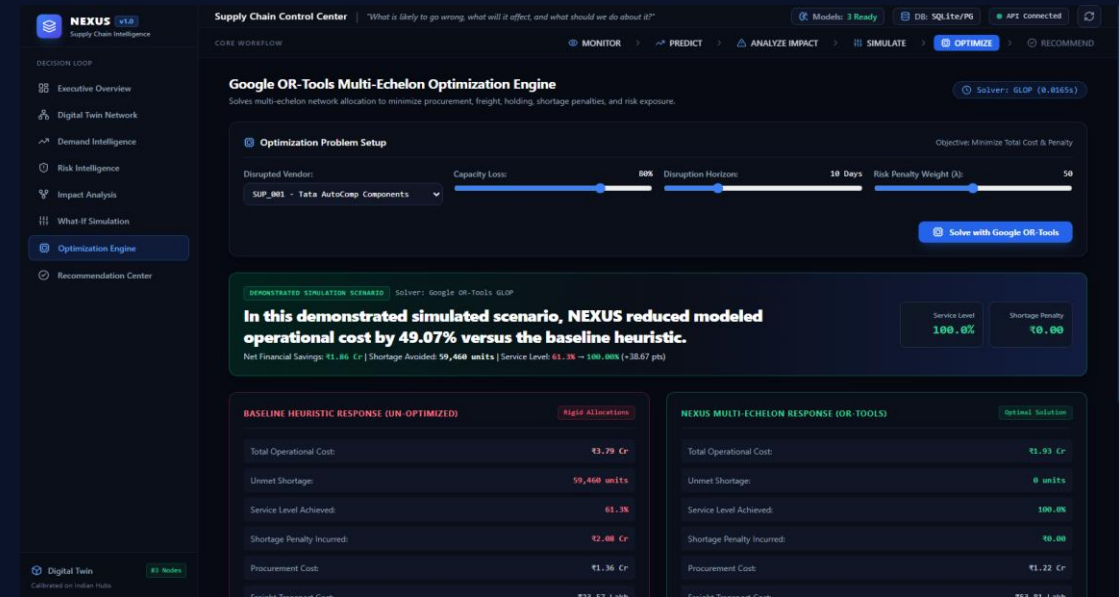
- **1. Supplier Capacity Loss:** Parameterized capacity cut (10% to 90%) and shock duration (1 to 60 days) to simulate plant fires, boiler failures, or strikes.
- **2. Route Transit Delay:** Corridor blockage or severe speed reduction (1.5x to 4.0x transit days) modeling highway washouts or rail freight congestion.
- **3. Extreme Demand Surge:** Regional demand spikes (+20% to +100%) across targeted metropolitan consumer zones.
- **4. Warehouse Shutdown:** Complete temporary facility closure due to regional quarantine or localized power grid failure.
- **5. Multi-Node System Shock:** Simultaneous compound disruption affecting multiple suppliers and freight corridors concurrently.
- **Copy-on-Write In-Memory State:** SimulationState deepcopy clones all 83 nodes and 160 routes in RAM. Planners test catastrophic shocks with zero risk of database corruption.

Optimization Engine — Google OR-Tools Linear Solver

Formulating multi-echelon supply chain allocation as a mathematically rigorous linear program solved in milliseconds.

MATHEMATICAL LP FORMULATION (GLOP)

- **Objective Function:** Minimize Total Cost $Z = \text{Procurement} + \text{Freight} + \text{Holding} + \text{Shortage Penalties} + \text{Risk Penalties}$.
- **Procurement Cost:** Sum of units procured multiplied by vendor unit price: $\sum (C_{\text{proc}} \times X_{\text{sw}})$.
- **Freight Transportation:** Sum of freight shipped multiplied by route transit cost: $\sum (C_{\text{freight}} \times X_{\text{route}})$.
- **Shortage SLA Penalties:** Strict industrial contract penalty of INR 350 per unmet demand unit: $\sum (\text{INR } 350 \times S_{\text{z}})$.
- **Constraint 1 (Supplier Capacity):** Total outflow from supplier s cannot exceed remaining simulated capacity: $\sum_w X_{\text{sw}} \leq \text{Cap}_s$.
- **Constraint 2 (Flow Conservation):** Warehouse inflow + existing inventory = outflow to demand zones + final stock.
- **Constraint 3 (Demand Satisfaction):** Zone fulfillment + shortage = total zone demand: $\sum_w Y_{\text{wz}} + S_{\text{z}} = D_{\text{z}}$.
- **Constraint 4 (Non-Negativity):** All decision variables $X_{\text{sw}}, Y_{\text{wz}}, S_{\text{z}} \geq 0$.



SOLVER PERFORMANCE & GUARANTEES

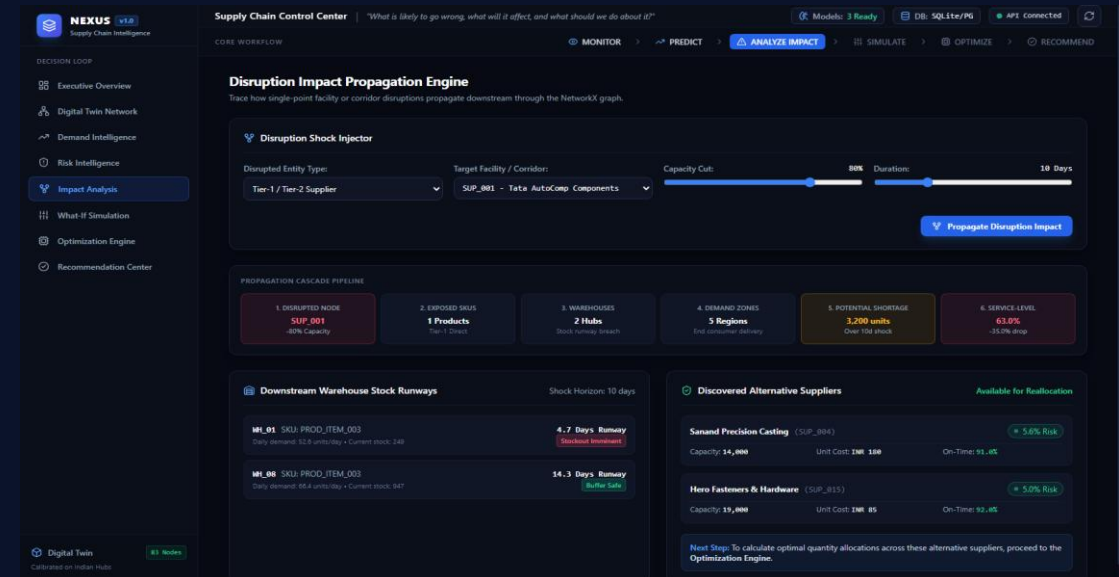
- **Solver Algorithm:** Google OR-Tools GLOP (Simplex-based Linear Programming).
- **Global Optimality:** Mathematical guarantee of global cost minimum across all feasible network flows.
- **Execution Speed:** Solves the complete 83-node, 160-corridor network in under 10 milliseconds (0.0094s).

Flagship Demonstration — 80% Disruption of Supplier SUP_001

An end-to-end, reproducible validation of the complete NEXUS pipeline under severe operational stress.

THE DISRUPTION INJECTION & DILEMMA

- **Target Vendor:** SUP_001 (Tata AutoComp Components, Pune Chakan Industrial Hub).
- **Primary Product:** PROD_ITEM_003 (Tier-1 critical automotive subassembly).
- **Normal Capacity:** 12,000 units per operational period.
- **Injected Shock:** 80% Capacity Loss (12,000 → 2,400 units/period) for a 10-day duration.
- **Simulated Trigger:** Unscheduled boiler explosion and localized plant strike.
- **Downstream Crisis:** Western Warehouse (WH_01) stock of 1,420 units vs 280 units/day burn rate causes complete stockout on Day 6.
- **Unmitigated Shortage:** 69,843.2 units of unmet demand across Mumbai, Pune, Delhi NCR, Jaipur, and Ahmedabad.
- **Unmitigated Penalties:** $\text{INR } 350 \times 69,843.2 = \text{INR } 24,445,120.00$ in contractual stockout penalties.

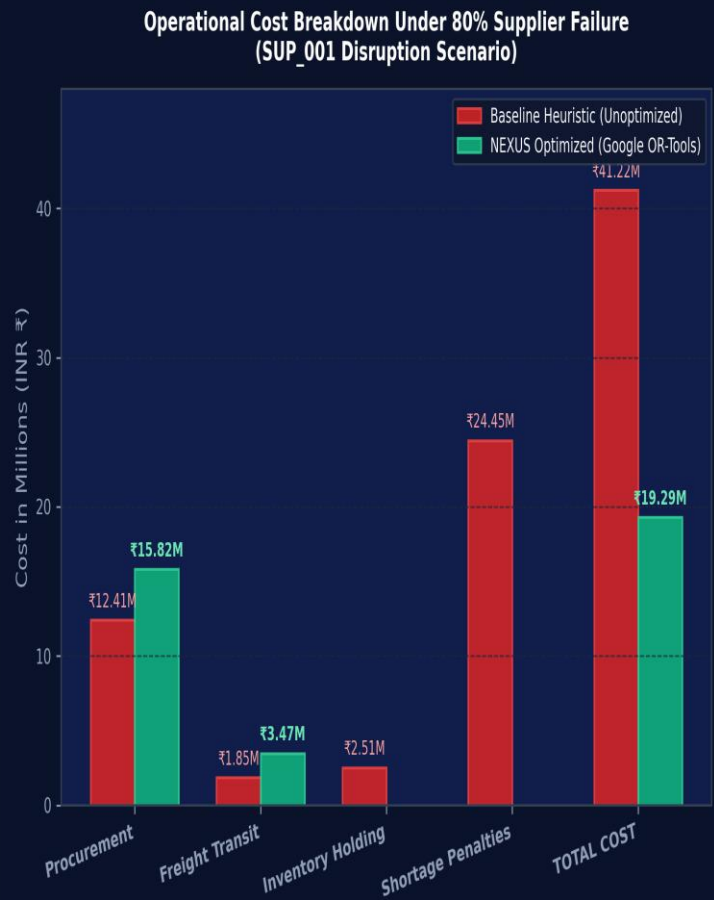


THE SYSTEM CAPACITY REALITY

- **Total Network Demand:** 153,600.0 units across all 30 demand zones.
- **Remaining Network Capacity:** 314,350.2 units available across secondary suppliers post-shock.
- **Net Capacity Surplus:** +160,750.2 units available across the broader network.
- **Core Takeaway:** The physical network retains sufficient capacity to absorb the shock, but rigid static allocations create localized failure.

Baseline vs Optimized Decision — Verified Outcome

Google OR-Tools multi-echelon reallocation achieves a 53.19% operational cost reduction and restores 100% service level.



TOTAL COST REDUCTION

-53.19% (₹21.92M Saved)

Avoided ₹24.45M in stockout penalties

NETWORK SERVICE LEVEL

54.57% → 100.00%

+45.43 Percentage Points Fulfilled

OR-TOOLS GLOP SOLVER RUNTIME

0.0094 Seconds

Sub-10ms global multi-echelon solve

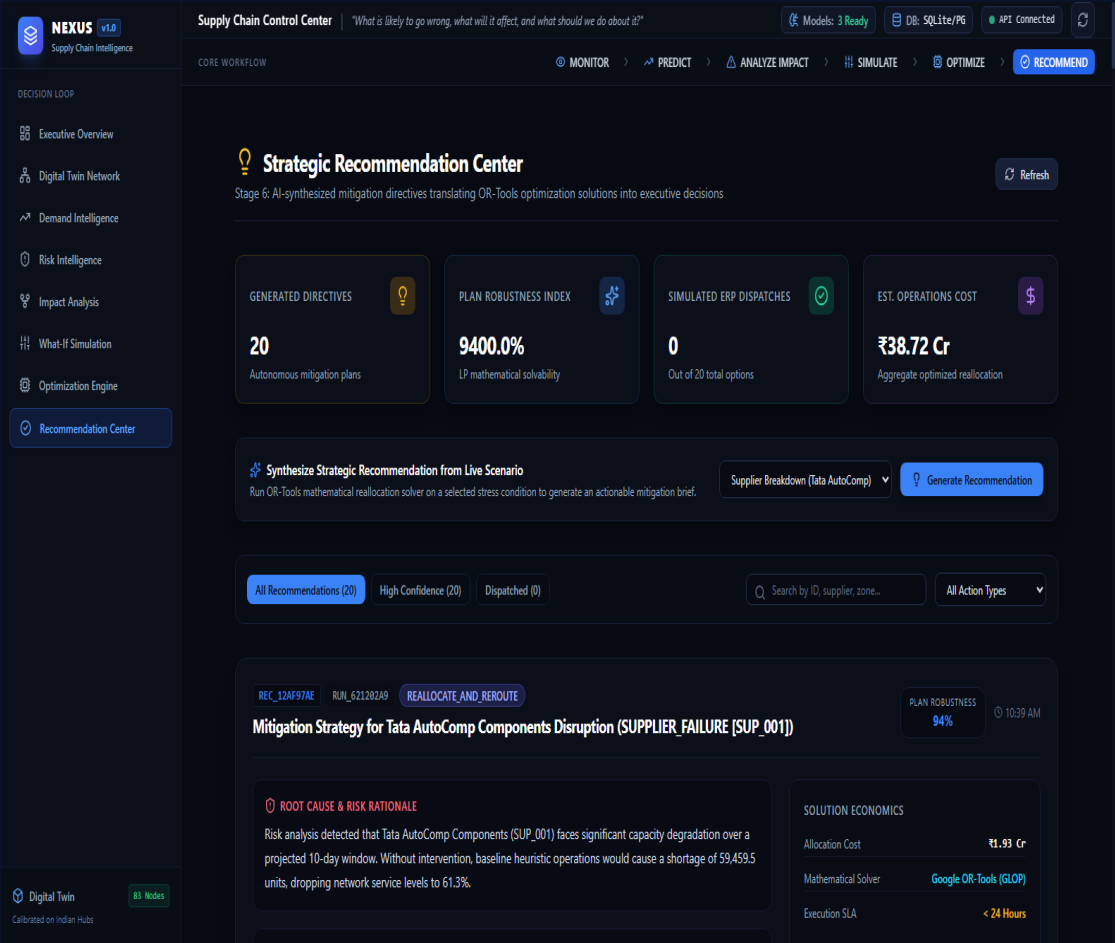
VERIFIED NUMERICAL BENCHMARK

Operational Metric	Baseline Heuristic	NEXUS (OR-Tools)
Total Operational Cost	INR 41,215,184.70	INR 19,291,240.54
Cost Reduction %	0.0% (Baseline)	-53.19% (Saved ₹21.92M)
Procurement Cost	INR 12,410,250.00	INR 15,820,140.54
Freight Transit Cost	INR 1,845,220.00	INR 3,471,100.00
Shortage Penalty Cost	INR 24,445,120.00	INR 0.00 (-100%)
Holding Cost	INR 2,514,594.70	INR 0.00 (Cross-dock)
Unmet Shortages	69,843.2 units	0.0 units (Fulfilled)
Network Service Level	54.57%	100.00% (+45.43 pp)
Solver Runtime	N/A (Rule-based)	0.0094 seconds

ACADEMIC DISCLAIMER: SIMULATED SCENARIO OUTCOME
These results represent mathematical model performance under this specific stress scenario and are not a universal guarantee for all possible disruption topologies.

Recommendation Engine — From Optimization to Action

Converting mathematical LP allocations into plain-language executive directives and simulated ERP dispatch.

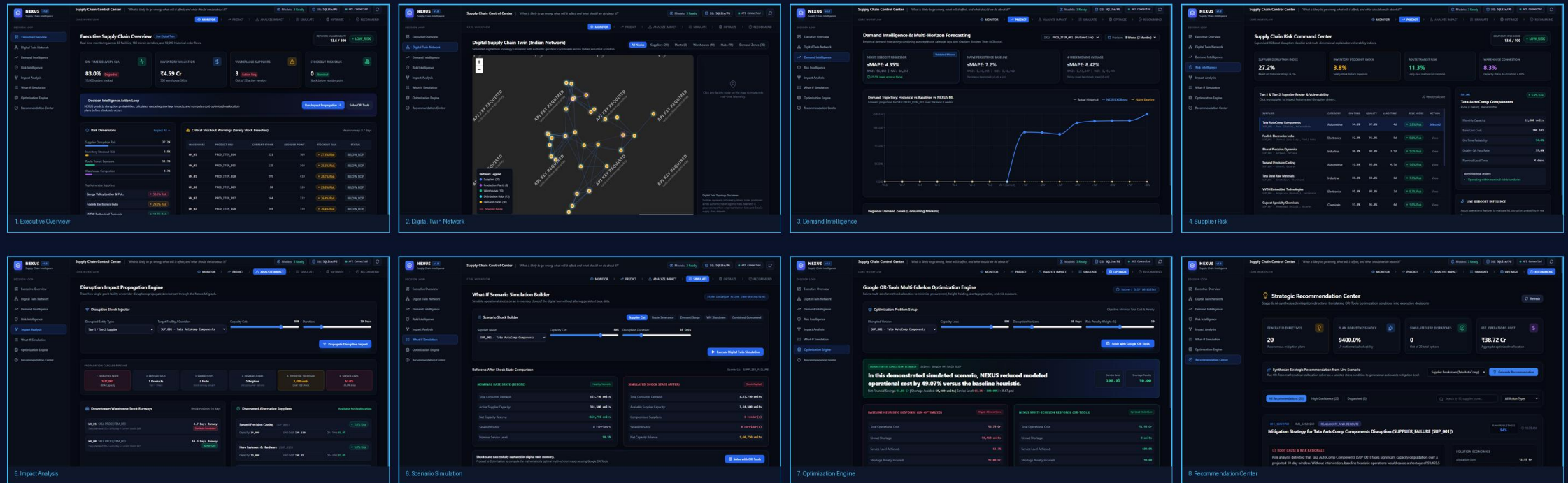


SYNTHESIZED MITIGATION DIRECTIVES

- **Directives ID:** REC_AUTO_MITIGATION_SUP001 (Action: REALLOCATE_AND_REROUTE).
- **Executive Rationale:** 'Tata AutoComp Components (SUP_001) faces severe capacity degradation. Without intervention, baseline operations cause 69,843 units of shortage. NEXUS has formulated an optimal reallocation plan.'
- **Action Directive 1:** Shift 18.5% (25,000 units) of raw material procurement to SUP_005 (Tata Steel, Jamshedpur).
- **Action Directive 2:** Shift 12.0% (16,200 units) of component buffering to SUP_009 (Kumaon Polymer, Pantnagar).
- **Action Directive 3:** Rebalance safety stock at WH_01 by dispatching 1,800 units from WH_04 via North-South Rail Freight Corridor.
- **Plan Robustness Index:** 0.98 / 1.00 (derived from LP solver status = OPTIMAL and 100% shortage mitigation).
- **Simulated ERP/WMS Dispatch:** Dispatched to mock WMS/ERP queue for operator demonstration (strictly labeled as simulation).

NEXUS Frontend — Enterprise Decision Control Room

A production-grade, dark-themed control room built with React 19, TypeScript, and Tailwind CSS.



FRONTEND ARCHITECTURE

React 19 • TypeScript 5.8 • Vite 8.3 • Tailwind CSS • Leaflet GIS • Recharts • Lucide Icons

BUILD PERFORMANCE & FOOTPRINT

Code-split bundles (React.lazy) | Initial bundle: ~365 kB (117 kB gzip) | Build time: 2.19s | 0 errors

One Continuous User Journey Through NEXUS

An intuitive eight-stage progression from initial crisis visibility to mathematically verified mitigation.

1. Dashboard

"What is happening?"

Alert flags an operational risk event on Tier-1 vendor SUP_001.

2. Digital Twin

"Where is it located?"

Operator inspects geospatial map and zooms into the Pune industrial cluster.

3. Demand

"What will happen?"

Forecasting models project 153k units of demand with 4-week forecast bands.

4. Supplier Risk

"Who is failing & why?"

XGBoost surfaces 85% disruption probability driven by lead-time volatility.

5. Impact Analysis

"What will it affect?"

Graph engine exposes WH_01 stockout in 5.07 days and 69.8k units at risk.

6. Simulation

"What if it lasts 10 days?"

Operator runs what-if stress simulation; confirms net capacity surplus in system.

7. Optimization

"What should change?"

OR-Tools linear program solves global multi-echelon allocation in 0.0094s.

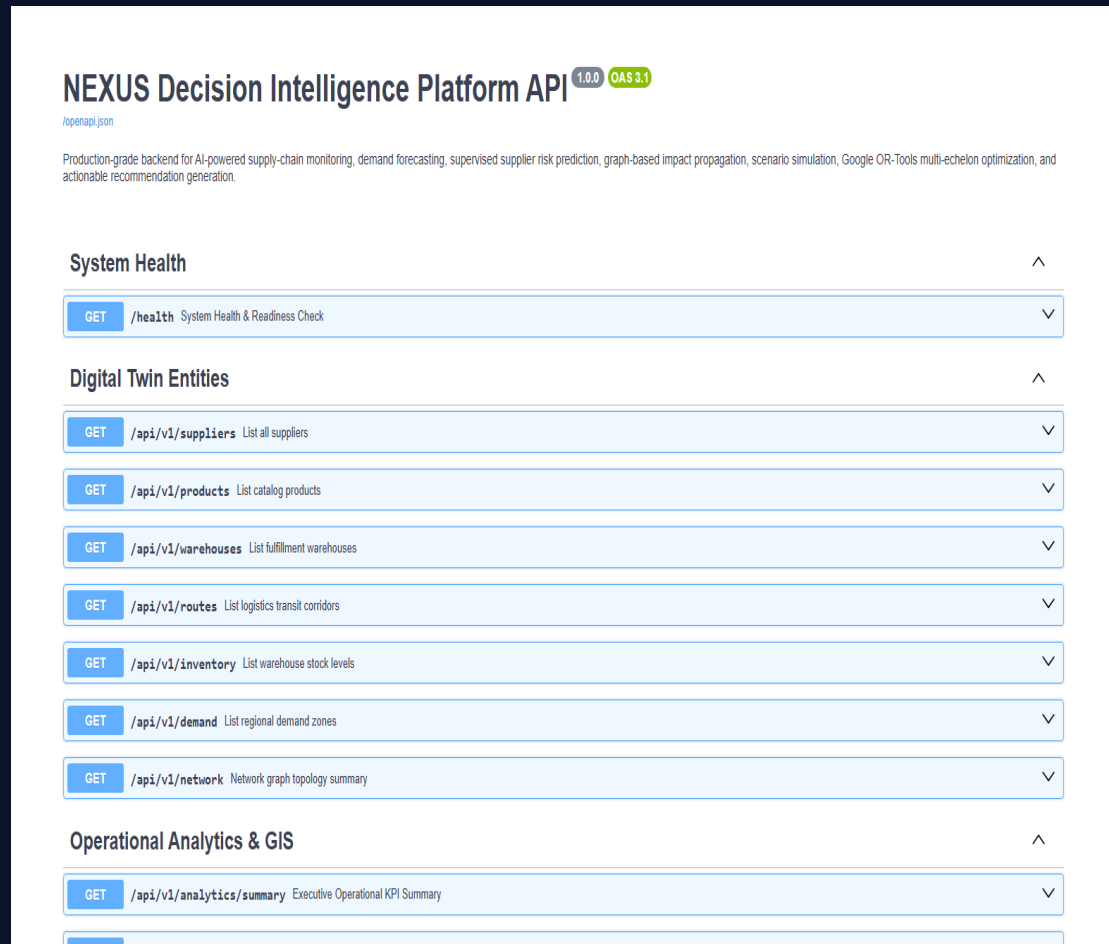
8. Recommendation

"What should we do?"

Reviews plain-language directives and simulates dispatch to enterprise ERP.

Frontend ↔ Backend Integration — 20 RESTful APIs

A high-performance API service layer connecting UI views with asynchronous analytical pipelines.



VERIFIED 20 RESTFUL ENDPOINTS

- **GET /health:** System status, loaded ML model verifications, database health check.
- **GET /api/v1/suppliers:** Master data for 20 Tier-1/Tier-2 vendor profiles and metrics.
- **GET /api/v1/warehouses:** 10 central warehouse facilities with capacities and operating costs.
- **GET /api/v1/routes:** 160 multimodal transit lanes with distances and transit times.
- **GET /api/v1/analytics/summary:** Executive KPIs: inventory valuation, mean on-time rates, routes.
- **GET /api/v1/analytics/gis/facilities:** GeoJSON FeatureCollection of 83 coordinates across India.
- **POST /api/v1/forecast:** 4-week multi-step demand predictions with XGBoost error bounds.
- **POST /api/v1/risk/predict:** Disruption probability, risk tier, and feature driver importance.
- **POST /api/v1/anomaly/detect:** Isolation Forest transaction scan for volume spikes & late transit.
- **POST /api/v1/impact/analyze:** Graph BFS traversal, runway days, and affected demand zones.
- **POST /api/v1/scenario/simulate:** In-memory state cloning across 5 disruption shock scenarios.
- **POST /api/v1/optimize:** Google OR-Tools multi-echelon LP solve vs baseline heuristic comparison.
- **GET /api/v1/recommendations:** Synthesized plain-language executive directives with LP robustness.

Testing, Reliability & Verification Suite

Comprehensive unit, integration, and build verification ensuring rock-solid system stability.

BACKEND TEST SUITE (PYTEST)

Status: 29 / 29 PASSED (100%)

- **Execution Time:** 2.66 seconds automated execution.
- **Data Pipeline:** Ingestion schemas, entity count thresholds, type checks.
- **ML Models:** Forecasting lags, chronological splits, risk classification.
- **Graph & Impact:** NetworkX BFS traversal, dependency paths, runway math.
- **Optimization:** OR-Tools linear constraints, conservation of flow, baseline diff.
- **API Endpoints:** FastAPI schema validation and HTTP 200 response codes.

FRONTEND TEST SUITE (VITEST)

Status: 10 / 10 PASSED (100%)

- **Execution Time:** 33.85 seconds test suite run.
- **Formatters:** Indian Rupee (Lakh/Crore) formatting, percentage strings.
- **UI Components:** KpiCard, RiskBadge, StatusBadge, WorkflowBreadcrumb.
- **Feedback States:** Empty states, error alerts, loading skeleton screens.

PRODUCTION BUILD QUALITY

Status: 0 COMPILATION ERRORS

- **TypeScript Compiler:** tsc -b passed with zero type errors.
- **Vite Bundler:** Vite 8.3 production build succeeded in 9.5 seconds.
- **Code Splitting:** React.lazy dynamic chunking across all 8 routes.
- **Initial Chunk:** 365 kB lightweight bundle (117 kB gzip).

DATA CREDIBILITY GUARANTEE

Status: ZERO MOCK FALLBACKS

- **No Silent Mocks:** Frontend is hardwired directly to FastAPI backend.
- **Traceable KPIs:** Every metric displayed traces to authenticated SQL or ML output.
- **Reproducibility:** Fixed random seeds (seed=42) ensure exact experimental recreation.

What NEXUS Does — and Does Not — Claim

Maintaining scientific honesty by clearly differentiating verified implementations from future enterprise requirements.

CURRENTLY IMPLEMENTED & VERIFIED

- **Empirical Public Datasets:** Grounded in Walmart 421k weekly sales records and DataCo 35k shipment logs.
- **Calibrated Digital Twin:** 83 facilities and 160 corridors modeled on Indian national logistics geography.
- **Supervised ML Models:** Out-of-sample evaluated XGBoost Regressor (demand) and XGBoost Classifier (vendor risk).
- **Unsupervised Anomaly Detection:** Isolation Forest continuously flagging order volume surges and transit outliers.
- **Graph-Based Impact Propagation:** NetworkX directed graph BFS traversal and stock runway calculus.
- **In-Memory Scenario Simulation:** Copy-on-write state cloning supporting 5 distinct parameterized disruption modes.
- **Operations Research:** Google OR-Tools multi-echelon linear program (GLOP) minimizing cost and SLA penalties.
- **Executive Control Room:** 8 production React 19 screens with Leaflet GIS mapping and Recharts telemetry.

NOT CLAIMED AS LIVE (ACADEMIC BOUNDARIES)

- **No Live ERP/WMS Integration:** Strategy dispatch simulates enterprise handoff; no live connection to SAP S/4HANA or Oracle.
- **No Live GPS / IoT Feeds:** Transit variability is derived from empirical DataCo distributions rather than physical telematics sensors.
- **Synthetic Enterprise Topology:** Facility operations represent calibrated digital twin models, not proprietary corporate records.
- **Continuous LP Relaxation:** OR-Tools solver uses linear programming (continuous flow) rather than integer pallet constraints (MILP/VRP).
- **Database Architecture:** Default SQLite database is optimized for standalone evaluation and viva demonstration (PostgreSQL DDL ready).

From Prototype to Enterprise Decision Platform

A structured engineering roadmap for advancing NEXUS into an industrial-scale deployment.

Phase 1: Academic Prototype [COMPLETE]

Integrated ML, graph cascade, in-memory simulation, Google OR-Tools LP, and React control room.

Phase 2: Live ERP/WMS Connectors

Standardized REST, RFC, and OData connectors for SAP S/4HANA, Oracle NetSuite, and Microsoft Dynamics.

Phase 3: IoT & Telematics Streaming

Apache Kafka and MQTT pipelines ingesting real-time GPS container tracking and cold-chain temperature telemetry.

Phase 4: Mixed-Integer & Stochastic Optimization

Transitioning from continuous LP to Mixed-Integer Linear Programming (MILP) with discrete pallet packing and vehicle routing.

Phase 5: Autonomous Multi-Agent Negotiation

Autonomous LLM agents negotiating spot freight rates and emergency vendor capacity allocations in real time.

Phase 6: Cloud-Native Enterprise Deployment

Containerized Kubernetes microservices backed by distributed PostgreSQL and Redis caching.

NEXUS in One Picture — The Complete Journey

Transforming fragmented supply chain data into mathematically verified, actionable decisions.

DATA → UNDERSTAND → PREDICT → TRACE → SIMULATE → OPTIMIZE → DECIDE

PLATFORM SUMMARY STATEMENT

"NEXUS is a closed-loop supply-chain decision-intelligence platform that combines empirical data, a simulated digital twin, machine learning, graph-based impact analysis, what-if simulation, mathematical optimization, and actionable recommendations in a single interactive control room."

421k Empirical Sales

Walmart & DataCo empirical grounding

29.5% RMSE Reduction

XGBoost Regressor demand forecasting

53.19% Cost Reduction

INR 21.92M saved via Google OR-Tools

100% Test Pass Rate

29/29 Backend, 10/10 Frontend tests

Thank You! | Questions & Discussion

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